PRODUCT INNOVATION HYPOTHESES

FORGE Innovation & Ventures

TARGET CUSTOMER: Warehouse and manufacturing enterprises using Autonomous Mobile Robots (AMRs), especially operations and safety managers.

FORGE INNOVATION HANDBOOK

HYPOTHESES

No. OF CUSTOMERS VALIDATED WITH

PROBLEM

Warehouse and manufacturing personnel operating in **high-risk and routine environments** lack **real-time situational awareness**, leading to **poor decision-making, safety risks**, and limited scalability of automation.

☐ 0 ☐ <5 ☐ <10 ☐ <20 ☐ >=20

CONSTRAINTS & BARRIERS

Constraints:
Industrial safety regulations, High accuracy required for navigation & sensing, Cost sensitivity for enterprise adoption, Integration with existing warehouse systems, Reliability in harsh industrial environments.
Barriers:
Resistance to replacing human monitoring , Trust gap between humans and autonomous robots , complexity of Deployment and setup , Skill gap in operating advanced robotic system .

☐ 0 ☐ <5 ☐ <10 ☐ <20 ☐ >=20

VALUE / BENEFITS

•Improved real-time situational awareness
•Faster and informed decision-making
•Safer warehouse and manufacturing operations
•Scalable automation with reduced human dependency

☐ 0 ☐ <5 ☐ <10 ☐ <20 ☐ >=20

SOLUTION & USABILITY

An **Autonomous Mobile Robot (AMR) platform** with AI-driven navigation and enhanced **Human-Robot Interaction**, enabling seamless monitoring, voice control, and efficient collaboration between humans and robots.

☐ 0 ☐ <5 ☐ <10 ☐ <20 ☐ >=20

PRICE

•**Subscription-based enterprise pricing**
•Lower upfront cost
•Includes maintenance, updates, and support

☐ 0 ☐ <5 ☐ <10 ☐ <20 ☐ >=20

DESIGNED BY : Forge Innovation & Ventures

Forge Innovation Rubric

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NECESSITY & SIGNIFICANCE	17/20	Real-world application scenarios	0 2 4
		Wide problem incidence	0 2 4
		High severity/impact of problem	0 3 6
		User/Customer validated use-cases	0 3 6
NOVELTY & ORIGINALITY	11/20	Unique & differentiated solution concept	0 3 6
		Holistic solution offering conceptually validated functionality	0 2 4 6
		(Futuristic) Potential to emerge as an enduring solution	0 4 8
CUSTOMER MOTIVATION	14/20	Customers have assigned highest priority & urgency	0 6
		Customers have failed with alternative solutions	0 2 4
		Customers have strong intent to co-create & standardise solution	0 2 6
		High degree of intent to invest in the immediate term	0 4
STRONG VALUE PROPOSITION	16/20	Demonstrate functional/operational benefits	0 3 6
		Significant gains measurable, quantified & validated	0 4 8
		Value proposition proven with reasonable number of customers	0 6
DEMONSTRATED PROOF OF VALUE	12/20	Field tested for utility, usability, integration & deployment constraints	0 4 6 8
		Strong indication of product adoption through successful field trails	0 3 6
		Proven for ease-of-adoption and accepted as permanent solution	0 3 6

70 /100

 2.0

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PROBLEM VALIDATION & CUSTOMER DISCOVERY

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PROBLEM - CUSTOMER - PRODUCT

Getting 'em all right!

List all the beneficiaries who would benefit or gain from solving this problem.

Select that beneficiary profiled in this canvas:

- [ ] Warehouse operator
- [ ] Manufacturing floor workers
- [ ] Safety and operation manager
- [x] Enterprise using AMR/ automation
- [ ]

Is this Beneficiary the RIGHT Customer?

Customer-Motivation	Low	Med	High
+ faces the problem most severely, and stands to benefit the most			✓
+ aware of the problem and able to set clear expectations on gains/benefits			✓
+ dissatisfied with existing solutions as they are complex, expensive, unusable, or ineffective			✓

Customer-Commitment	Low	Med	High
+ willing to describe solutions attempted and specify deployment/ usability constraints		✓	
+ strong intent to co-create an effective solution		✓	
+ willing to define the evaluation/acceptance criteria for buying the product			✓
+ indicates the preferred price range for buying a product		✓	

PV-CD Progress Indicator

No of Customers	0	<5	<10	<20	>=20
Problem definition documented as is being experienced by the beneficiary					✓
Magnitude of the problem (PAIN) has been measured and validated with beneficiary			✓		
Incidence/Occurrence of the problem (FREQUENCY) has been assessed and validated with beneficiary					✓
Challenges/limitations of current solutions attempted have been discussed/analysed			✓		
Gaps/Challenges with existing commercial products have been analysed/documented			✓		
Adoption barriers have been discussed/documented	✓				

01 Problem Description

Can you describe the problem in your own words exactly how you experience it?

In warehouse and manufacturing environments, there is no proper real-time visibility and interaction between humans and autonomous robots, making operations unsafe and inefficient.

02 Tried Solutions

How have you tried to solve this problem before? Can you explain the solution?

By using manual supervision, basic autonomous robots, and static monitoring dashboards.

03 Gaps

Why did you not make that solution permanent? What were the gaps in the solution?

The solutions lacked real-time awareness, had poor human–robot interaction, required continuous human involvement, and were not scalable.

04 Constraints/Barriers

What are the other constraints/challenges that have to be overcome or accommodated while solving the problem?

Safety regulations, integration with existing systems, accuracy requirements, deployment complexity, and cost constraints.

05 Benefits

How will you benefit from solving the problem? How much is solving this problem worth to you?

It improves safety, reduces human effort, increases efficiency, and is worth investing in due to long-term operational cost savings.

06 Importance

How important to you is solving this problem among other priorities?

High Importance

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FORGE MUP CHALLENGE BRIEF

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CHALLENGE STATEMENT

Describe the challenge statement as it is faced in the real-world by the target beneficiaries.

In real warehouse and manufacturing environments, operators struggle to safely and efficiently manage autonomous robots due to the lack of real-time situational awareness and effective human-robot interaction, which leads to delayed decisions, safety risks, and inefficient operations.

GAP ANALYSIS

Alternatives

How does the target user solve the challenge today?

By manual supervision, basic AMRs, and static monitoring dashboards.

What are the other alternatives available in the market today?

Standard AMRs with limited interfaces and conventional warehouse automation systems.

Gaps

What are the gaps in the way the challenge is solved today?

Lack of real-time awareness, poor human-robot interaction, and high dependence on manual monitoring.

What are the gaps/shortcomings?

Delayed information, limited scalability, complex operation, and safety limitations.

CONSTRAINTS

What are the usability/ deployment constraints to be considered for the solution to be easy to use and to adopt?

Usability

Skills/Expertise
Time Consumption
Physical/Personal Risk
Product Training
Changes to habits/Process
Accessibility
OTHERS:

The solution must be easy to learn, require minimal training, integrate smoothly with existing systems, comply with safety regulations, operate reliably in industrial environments, and remain cost-effective for enterprise adoption.

Deployment

Total Cost of Ownership
Maintenance/Service
Installation/Integration
Resources/Materials
Time Consumption
OTHERS:

USE-CASE

From the different scenarios in which the challenge is faced, describe the specific use-case you wish to target.

Real-time monitoring and interaction with autonomous mobile robots during routine and high-risk operations in warehouse and manufacturing environments.

TARGET USER

For this use-case describe the profile of the target user that you expect to adopt/test and accept your MUP as a permanent solution.

Real-time monitoring and interaction with autonomous mobile robots during routine and high-risk operations in warehouse and manufacturing environments.

UTILITY/FEATURES

Jobs

Describe the specific job the target user wants to complete (get done)?

To safely and efficiently monitor, interact with, and manage autonomous robots in real time during warehouse and manufacturing operations.

Outcomes

What are the expected outcomes? How do you quantify them?

Improved safety (reduced incidents), faster decision-making (reduced response time), and higher efficiency (increased robot uptime and task completion rate).

Pains

What are the pains to be relieved?

Safety risks, delayed decision-making, poor real-time visibility, and heavy dependence on manual monitoring.

Gains

What are the gains to be created?

Safer operations, better control over robots, reduced human effort, and scalable automation.

REFERENCES

Indicate other details or reference materials to know more about the user/use-case and the existing solutions/alternatives

Industry reports on warehouse automation, AMR vendor documentation, case studies of AMR deployments, safety standards for industrial robotics, and existing AMR product datasheets and demos.

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VPC ACTIVITY GUIDE

Startup Name:	Date:	Version:
Team Members:		

FORGE INNOVATION HANDBOOK

Pains to be relieved

Cost: What does your customer find too costly? (takes a lot of time, costs too much money, requires substantial efforts)

Feeling Bad: What makes your customer feel bad? (frustrations, annoyances, things that give them a headache)

Performance of Known Competition: How are current solutions underperforming for your customer? (lack of features, performance, malfunctioning)

Main Difficulties & Challenges: What are the main difficulties and challenges your customer encounters? (understanding how things work, difficulties getting things done, resistance)

Negative social consequences: What negative social consequences does your customer encounter or fear? (loss of face, power, trust, or status)

Risks: What risks does your customer fear? (financial, social, technical risks, or what could go awfully wrong)

Keeps Awake at Night: What's keeping your customer awake at night? (big issues, concerns, worries)

Common Mistakes: What common mistakes does your customer make? (usage mistakes)

Barriers: What barriers are keeping your customer from adopting solutions? (upfront investment costs, learning curve, resistance to change)

Customers Jobs-to-be-done

Functional Job (Perform, complete a specific task, solve a specific challenge)

Safely monitor and manage autonomous robots in real time during operations.

Social Job (Gain power, looking good or status)

Demonstrate advanced automation capability and operational excellence.

Emotional Job (aesthetics, feel good, security)

Feel confident, secure, and in control of automated systems.

Basic Needs (Communication, nutrition, belonging, etc)

Clear communication, safety assurance, and reliable coordination between humans and robots.

Gains to be created

Savings: Which savings would make your customer happy? (time money and effort)

Outcome Expectations: What outcomes does your customer expect and what would go beyond his/her expectations? (quality level, more of something, less of something)

Current Solutions – Delight?: How do current solutions delight your customer? (specific features, performance, quality)

Life or Job – Easier?: What would make your customer's job or life easier? (flatter learning curve, more services, lower cost of ownership)

Positive Social Consequences: What positive social consequences does your customer desire? (makes them look good, increase in power, status)

What are customers looking for? (e.g. good design, guarantees, specific or more features

What do customers dream about? (e.g. big achievements, big reliefs)

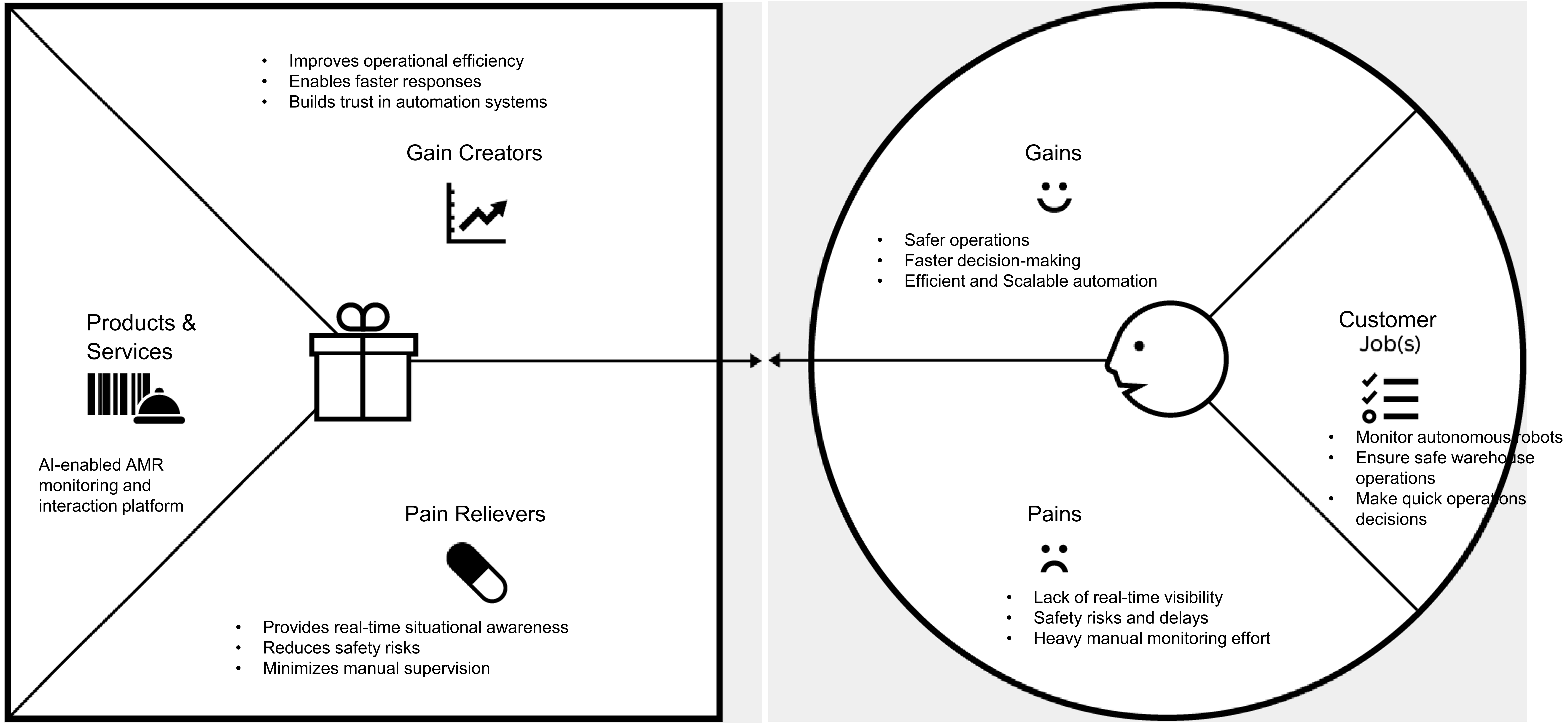
How does your customer measure success and failure? (e.g. performance, cost)

What would increase the likelihood of adopting a solution? (e.g. lower cost, less investments, lower risk, better quality, performance, design)

The Value Proposition Canvas

Value Proposition

Customer Segment



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VPC ACTIVITY GUIDE

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Our value proposition is for (Target user)

Our value proposition is for **warehouse and manufacturing enterprises using autonomous robots**.

who (Challenge statement)

Who **struggle to safely and efficiently monitor and interact with autonomous robots due to lack of real-time situational awareness**.

the (Solution)

The solution is an **AI-enabled autonomous robot platform that provides real-time situational awareness and seamless human–robot interaction for safer and more efficient operations**.

is a (Product category/theme)

Is an **AI-driven Autonomous Mobile Robot (AMR) and Human–Robot Interaction platform**.

it has (Compelling reason to buy)

It has **real-time visibility, improved safety, faster decision-making, and reduced human effort**.

unlike (Primary competitive alternatives)

Unlike **manual supervision, basic AMRs, and static monitoring dashboards**.

our Product (Primary differentiation)

Our product **enables real-time situational awareness with intuitive human–robot interaction, making automation safer and easier to manage**.

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MUP CONCEPTS GENERATION

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VALUE PROPOSITION

What are your Products & Services?
What are your Gain Creators?
What are your Pain Relievers?

Products & Services:
AI-enabled autonomous robot platform with real-time monitoring and human–robot interaction.
Gain Creators:
Faster decisions, improved efficiency, safer operations, scalable automation.
Pain Relievers:
Reduces safety risks, manual monitoring effort, delayed information, and operational inefficiencies.

ADOPTION BARRIERS

What are the constrains/challenges that have to be overcome or accommodated while solving the problem?

Compliance with safety regulations, smooth integration with existing systems, accuracy and reliability requirements, user training needs, and cost constraints.

CONCEPT LEGOS

Concept Generation

TOOLS/TECHNIQUES

FUNCTIONS	Option #1	Option #2	Option #3	Option #4	Option #5
Real-time Monitoring	Live robot status dashboard	Camera-based visual monitoring	Sensor data visualization (LiDAR/GPS)	Mobile app monitoring	Cloud-based monitoring system
Human–Robot Interaction	Manual joystick control	Touchscreen interface	Voice command interaction	Gesture-based control	AI conversational interface
Safety & Risk Handling	Emergency stop button	Obstacle detection sensors	Collision avoidance algorithm	Safety alert notifications	Predictive risk analysis
Navigation & Movement	Predefined path navigation	GPS-based navigation	SLAM-based autonomous navigation	Assisted manual navigation	Fully autonomous adaptive navigation
Decision Support	Basic status indicators	Rule-based alerts	Data-driven recommendations	AI-assisted decision prompts	Predictive decision intelligence

MUP CONCEPTS ASSESEMENT

FUNCTIONS	CONCEPT #__	CONCEPT #__	CONCEPT #__
• Real-time robot monitoring • Human-Robot interation • Safety & risk handling • Automation navigation • Decision support	**Monitoring Dashboard–Based AMR Control** **Capability (how feasible to build with current capabilities):** High, as it uses existing sensors, dashboards, and known software tools. **Usability (overcoming adoption barriers):** High, because operators can easily learn and use visual dashboards. **Feasibility (time & cost to build prototype):** High, since it requires moderate development time and reasonable cost.	**Voice & AI Interaction–Enabled AMR** **Capability (how feasible to build with current capabilities):** Medium–High, as voice interfaces and AI models are available but need integration. **Usability (overcoming adoption barriers):** High, as hands-free control improves ease of use and operator acceptance. **Feasibility (time & cost to build prototype):** Medium, due to additional development and testing effort.	**Fully Autonomous AMR with Predictive Intelligence** **Capability (how feasible to build with current capabilities):** Medium, as advanced AI and autonomy require specialized expertise. **Usability (overcoming adoption barriers):** Medium, because trust and training are required for full autonomy. **Feasibility (time & cost to build prototype):** Low–Medium, as it needs higher time, cost, and validation effort.
Capability How feasible is the concept to build considering your capabilities? **Usability** How usable is the prototype overcoming adoption barriers to test the value proposition? **Feasiblity** How feasible is the solution to build considering the time and cost in building the prototype?	0 - 10 Capability: 8 Usability: 7 Feasiblity: 8	0 - 10 Capability: 7 Usability: 8 Feasiblity: 6	0 - 10 Capability: 6 Usability: 7 Feasiblity: 5

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MUP (Tech Canvas)

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USER/USE CASE

Real-time monitoring and interaction with Autonomous Mobile Robots in warehouse and manufacturing environments.

PIR:

Lack of real-time situational awareness and effective human–robot interaction causes safety risks and delayed decisions.

IQ:

How can humans safely and efficiently monitor and interact with autonomous robots in real time?

VALUE PROPOSITION:

Provides real-time visibility and seamless human–robot interaction to improve safety, efficiency, and scalability of automation.

FEATURES AND FUNCTIONALITIES:

- Real-time robot monitoring
- Human-robot interaction (voice/dashboard)
- Safety alerts and risk detection
- Autonomous navigation support

ACTORS AND ROLES:

Warehouse operator – monitors and controls robots
Safety manager – ensures safe operations
AMR system – executes autonomous tasks

UI/UX:

SYSTEM CONTEXT:

Industrial warehouse and manufacturing environments with autonomous robots, sensors, and human operators.

SYSTEM DESIGN:

SYSTEM CONSTRAINTS:

Safety regulations, integration with existing systems, accuracy and reliability requirements, cost and infrastructure limitations.

INPUT:

PHYSICAL QUANTITY:

- Distance
- Position
- Obstacles

SENSOR/TRANSDUCER:

LiDAR, camera, GPS, proximity sensors

INTERFACE :

Wired / wireless sensor interfaces

COMPUTE: HARD DISK MEMORY:

Onboard processor / edge computer

POWER:

Battery-powered system

INTERFACE :

Wired / wireless sensor interfaces

OUTPUT

ACTUATOR :

Motors, brakes, alert systems

PHYSICAL QUANTITY:

Robot movement, alerts, status signals

HMI:

Monitoring dashboard / voice interaction interface.

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PRODUCT CANVAS

Startup Name: | Date: | Version:

Team Members:

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PROBLEM STATEMENT 01

Which problem do you aim to solve?

*Describe the problem as it is faced in the real world. If you are solving a technical problem then find a real-world product/service where this tech is currently applied, that's a good starting point.

Why is solving this problem very important?

Who are the different beneficiaries?

What goal can be achieved by solving this problem?

What gaps/challenges - in the way the problem is solved today, have to be overcome?

What are the different real-world scenarios where this problem is faced most critically?

In warehouse and manufacturing environments, operators lack real-time visibility and effective interaction with autonomous robots, leading to safety risks, delayed decisions, and inefficient automation.

TARGET CUSTOMER 02

Which beneficiary (type/category) gains most from solving this problem?

What should we look for if we have to find real-world customers of this beneficiary (type/category) in the real-world?

What does this beneficiary (type/category) expect as benefits/outcomes?

What would it be worth for this beneficiary (type/category) to solve this problem?

Warehouse and manufacturing enterprises using AMRs, especially operations managers and safety teams responsible for monitoring and controlling robots.

PRODUCT 03

How are you solving this problem for your target customer?

What constraints faced by this customer are you overcoming with your solution?
*costs, resources, skills, operating conditions, time, convenience

How do you make this solution easy to adopt and use for your target customer?

An AI-enabled autonomous robot platform that provides real-time monitoring, situational awareness, and seamless human–robot interaction.

VALUE PROPOSITION 04

Does your product help your target customer make or save money?

How is your product better than the alternative available today?

What extra benefits does the customer get from your product?

How do you make it easy for the customer to measure or experience the benefits?

An AI-enabled autonomous robot platform that provides real-time monitoring, situational awareness, and seamless human–robot interaction.

UNFAIR ADVANTAGE 05

In what other ways can your customer achieve the goal?

What other offerings in the market does your target customer compare your solution with now or is likely to do so in the future?

Who are your most important present and future competitors?

What serious business advantages do you enjoy over the competitors

What serious tech, operating or business advantages do you enjoy over your current or future competitors?

Integrated real-time awareness and human–robot interaction in a single platform, tailored for industrial environments and easy adoption.

CUSTOMER ACQUISITION & SALES 06

How will you create awareness and generate leads for your product?

How do you increase the conversion from leads to trials, and from trials to sales?

Where would your customers actually buy your product from?

What is the channel for distribution and delivery of your product to the customer?

Direct sales to enterprises, pilot deployments, demos in warehouses, and partnerships with automation solution providers.

PRICE & REVENUE MODEL 07

Who pays for your product? Who is the buyer & user? Are they the same?

For what value is your target customer willing to pay?

How much is your solution worth to your target customer?

Are other sources of revenue possible from other beneficiaries?

Subscription-based pricing with optional customization and maintenance services for enterprises.

METRICS 08

Which outcomes should you measure to validate the assumptions in the creation & delivery of your core value proposition?

Are you maximising the conversion rate in the Get-Keep-Grow pipeline for your target customers?

Reduction in safety incidents, faster response time, improved robot uptime, and increased operational efficiency.